

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence
NAME: R. RUPA SREE

COURSE CODE: CSA17
REGISTER NUMBER: 192472122

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration: 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I R. Rupa Sree (192472122) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

R. Rupa Sree

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning*, *Decision Trees*, *Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM)*, or *Reinforcement Learning applications*.

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

ARTICLE REVIEW

Article: Real-world validation of safe reinforcement learning, model predictive control and decision tree-based home energy management systems.

Article Citation Details

Ruddick, J., Ceusters, G., Van Krieking, G., Genov, E., De Cauwer, C., Coosemans, T., & Messagie, M. (2024). Real-world validation of safe reinforcement learning, model predictive control and decision tree-based home energy management systems. *Energy and AI*, 18, Article 100448. <https://doi.org/10.1016/j.egyai.2024.100448>

Publisher: Elsevier

Journal: *Energy and AI*

Year: 2024

Volume: 18

Article number: 100448

DOI: 10.1016/j.egyai.2024.100448

Task 1: Article Summary

1(a) Domain and ML Problem

The article focuses on intelligent home energy management, an application area combining artificial intelligence, renewable energy, electric vehicles and battery energy storage. The specific machine-learning problem is how to control household energy resources safely and economically under practical operating constraints. The study evaluates a reinforcement-learning method equipped with a safety layer and a decision-tree control policy, and compares them with model predictive control and simple rule-based control. The central problem is not only reducing energy cost but also ensuring that learned control actions remain safe when deployed in a real-world electrical system.

1(b) Objective, Research Gap and Proposed Solution

The main objective of the article is to validate machine-learning-based home energy-management methods in a real-world experimental environment rather than relying only on computer simulations. Previous studies had shown promising results for reinforcement learning and decision-tree-based control, but their effectiveness had mainly been demonstrated in simulation. This created a practical research gap: an algorithm that performs well in simulation may behave differently when exposed to measurement errors, physical constraints, changing loads and real electrical equipment. To address this gap, the authors implemented a safe reinforcement-learning approach using an OptLayerPolicy safety layer and a decision-tree control approach called TreeC. They compared these methods with model predictive control and simple rule-based control. The experiments used four reproductions of residential houses containing batteries, photovoltaic systems, dynamic electrical loads and controllable electric-vehicle chargers. The study therefore evaluates both economic performance and operational safety. The work is important because it moves the evaluation of AI-based energy management from purely simulated environments toward realistic hardware-based testing.

Task 2: Methodology Analysis

2(a) ML Techniques and Experimental Data

The paper investigates two main machine-learning approaches. First, safe reinforcement learning uses an OptLayerPolicy safety layer to constrain the actions of an RL agent during online operation. The agent learns a control policy through interaction with the energy-management environment while the safety layer helps prevent unsafe actions. Second, TreeC generates an interpretable decision-tree control policy using a metaheuristic approach. The resulting tree can be inspected and used as a control strategy. For comparison, the authors also use model predictive control and simple rule-based control as benchmark methods. The experimental platform consists of four reproductions of residential houses. Each setup includes a battery energy-storage system, photovoltaic generation, dynamic electrical load representing non-controllable consumption, and a controllable electric-vehicle charger. The study is therefore based on real electrical installations and hardware-in-the-loop experimentation rather than a conventional classification dataset. Inputs include measurements and operating information needed by the energy-management controllers, while the evaluation focuses mainly on electricity cost and constraint violations. The authors also perform additional simulations to investigate the effect of a more representative training dataset for TreeC and implementation issues affecting model predictive control.

2(b) Mapping to CO4

Topic	Connection with Article	Evidence/Role
Reinforcement Learning	Safe RL / OptLayerPolicy	Agent learns energy-management actions through interaction while a safety layer handles constraints.
Decision Trees	TreeC control policy	A decision-tree policy is generated from data and used for interpretable control.
Inductive Learning	Learning control decisions from observations	The TreeC policy derives decision rules from observed/training information.
Statistical Learning	Not the primary technique	The paper is mainly focused on RL and decision-tree control rather than Naïve Bayes, Logistic Regression or SVM.

Methodological Strength

A major strength is real-world validation. The authors do not stop at simulation results; they test the methods on four residential-house reproductions containing physical energy-management components. This makes the evaluation more practically meaningful and exposes issues that may not appear in simulation.

Methodological Limitation

A key limitation is the relatively small experimental scope: only four residential-house reproductions are used. Also, the reinforcement-learning method is still in its training phase during the reported comparison, making the comparison less favourable to RL and preventing a definitive conclusion about its final performance. The safety layer is also reported to remain error-prone.

Task 3: Results and Critical Evaluation

3(a) Key Results

Method	Reported result	Interpretation
Simple rule-based control	Similar cost to TreeC and MPC	Strong baseline despite its simplicity.
TreeC decision-tree control	Within the group achieving similar costs	Provided the safest operational performance in the reported comparison.
Model Predictive Control	Similar cost to rules and TreeC	Competitive but dependent on accurate data from multiple sources.
Safe Reinforcement Learning	25.5% higher cost; still training	Shows promise for safe online learning but had not yet reached comparable economic performance.
TreeC safety performance	27.1 Wh grid-limit exceedance	Lowest reported exceedance among the highlighted comparison.
RL safety performance	593.9 Wh grid-limit exceedance	Higher reported exceedance than TreeC.

The article reports that simple rules, TreeC and model predictive control achieved costs within a difference of only 0.6%, whereas the reinforcement-learning method, which was still training, had a cost 25.5% higher. The authors also report grid-limit exceedances of 27.1 Wh for TreeC compared with 593.9 Wh for reinforcement learning.

3(b) Critical Evaluation

The reported results are realistic because they come from real experimental electrical installations rather than only a synthetic benchmark. However, the results should not be interpreted as proof that decision trees are always better than reinforcement learning. The RL method was still in training, so it may not represent its converged performance. At the same time, the experiment uses only four residential-house reproductions, which limits statistical generalization to different buildings, climates, household behaviours and energy tariffs. Potential bias can arise from the selection of houses, the training data, the operating conditions and the evaluation period. The authors themselves identify the importance of a representative training dataset for TreeC. Another concern is that different methods may have different levels of maturity or tuning, so a fair comparison should use comparable development effort and evaluation conditions. Additional experiments would strengthen the findings. These should include a larger number of houses, multiple seasons, different electricity-price structures, different photovoltaic and battery sizes, longer RL training until convergence, repeated trials with different random seeds, and statistical confidence interval. Testing the controllers under sensor noise, communication failures and unexpected loads would provide stronger evidence about industrial robustness.

3(c) Real-World Applicability

The approach can be deployed in smart homes, residential microgrids, building energy-management systems, electric-vehicle charging systems and distributed battery-storage systems. Decision-tree policies are attractive where interpretability is important, while safe RL could be useful where operating conditions change and an adaptive controller is required. Scaling to industry creates several challenges. Large-scale deployment requires reliable real-time data from smart meters, batteries, photovoltaic systems and EV chargers. Models must handle changing household behaviour, equipment differences and electricity-price changes. RL also requires sufficient training and safe exploration, which is difficult when actions can affect physical equipment. Cybersecurity, privacy, reliability, computational cost and safety certification are additional concerns. Therefore, a practical industrial system would benefit from safety constraints, monitoring, fallback controllers and extensive validation before autonomous operation.

Task 4: Reflection and Learning Outcome

4(a) Three Key Insights

1. Reinforcement learning is useful for sequential decision-making. The article showed that an RL controller can learn energy-management behaviour through interaction with an environment. This deepens the CO4 concept of Reinforcement Learning, where an agent learns actions based on states, actions and rewards.
2. Interpretability can be important in machine learning. The TreeC approach produces a decision-tree control strategy that can be inspected and understood more easily than a complex learned policy. This connects to the CO4 topic of Decision Trees and shows why interpretability matters in safety-sensitive applications.
3. Real-world evaluation is different from simulation. A method can appear successful in a simulated environment but face measurement errors, physical constraints and unexpected behaviour during deployment. This insight connects CO4 machine-learning implementation with model evaluation and practical decision-making.

4(b) Proposed Novel Extension

A useful extension would be a hybrid Safe-RL + Decision-Tree system evaluated across a much larger and more diverse collection of residential buildings. The RL agent could learn an adaptive policy, while a decision-tree safety or fallback layer could constrain actions and provide an interpretable emergency policy. The experiment could compare the hybrid system against standalone RL, TreeC, MPC and rule-based control under different seasons, electricity prices, weather conditions and household demand patterns.

This experiment is feasible because the existing study already demonstrates real-world testing and provides code/data resources. A hybrid approach could combine the adaptability of RL with the interpretability and operational safety of decision-tree control. The expected impact would be improved robustness, safer deployment and potentially lower energy cost while retaining a transparent fallback mechanism.

Overall Conclusion

The reviewed article provides a valuable CO4 example because it connects reinforcement learning, decision trees and practical machine-learning-based decision-making. Its most important contribution is the real-world validation of methods that had previously shown promise mainly in simulation. The results demonstrate that TreeC, simple rules and MPC achieved very similar costs, while the RL method had higher cost because it was still training. The study also highlights that safety and reliability are as important as optimization performance when machine learning is deployed in physical systems. Overall, the article shows that successful AI deployment requires not only a strong algorithm but also realistic evaluation, safety mechanisms, representative data and careful consideration of real-world constraints.

Reference

Ruddick, J., Ceusters, G., Van Krieking, G., Genov, E., De Cauwer, C., Coosemans, T., & Messagie, M. (2024). Real-world validation of safe reinforcement learning, model predictive control and decision tree-based home energy management systems. *Energy and AI*, 18, 100448. <https://doi.org/10.1016/j.egyai.2024.100448>

RUBRICS FOR CALCULATION:

Criteria	Weight age (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand